REPUBLIC OF MOZAMBIQUE

May 11, 2012

FOURTH REVIEW UNDER THE POLICY SUPPORT INSTRUMENT AND REQUEST FOR MODIFICATION OF ASSESSMENT CRITERIA—DEBT SUSTAINABILITY ANALYSIS

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This debt sustainability analysis (DSA) updates the joint IMF/IDA DSA from May 24, 2011 reflecting the most recent macroeconomic developments.[9] Consistent with the previous DSA, its main results indicate that Mozambique remains at a low risk of debt distress as the government plans to temporarily increase public investment partially financed by external borrowing on nonconcessional terms over the next few years. However, further improvements in debt management and project selection capacity will be key to use nonconcessional resources productively.[10] As public debt is largely external, the evolution of public debt indicators mirrors that of external debt.

[9]In line with the 2010 Staff Guidance Note, a full joint LIC DSAs is expected to be prepared once every three years for PRGT-eligible IDA-only countries. In between, short annual updates are expected to be produced unless macroeconomic conditions since the last full DSA have significantly changed. See *Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries* (www.imf.org) and IDA/SECM2010-0029.

[10]The DSA presented in this document is based on the standard low-income countries (LIC) DSA framework. See *Debt Sustainability in Low-Income Countries: Further Considerations on an Operational Framework, Policy Implications* (www.imf.org) and IDA/SECM2004/0629. Under the Country Policy and Institutional Assessment (CPIA), Mozambique is rated as a medium performer, with an average rating of 3.71 during 2008–10, and the DSA uses the indicative threshold for countries in this category.

I. UNDERLYING DSA ASSUMPTIONS

5. This DSA is consistent with the macroeconomic framework outlined in the Staff Report for the Fourth Review under the Policy Support Instrument. Compared to the previous DSA,[11] this update includes:

a. **A revised medium-term macroeconomic framework.** It takes into account a temporary, albeit small, dent in economic growth in 2012 as a result of the global downturn, but a slightly improved outlook over the medium to long term as a result of booming investment and production in the mining and hydrocarbon sectors. The analysis also assumes, like in the past, that Mozambique will continue its prudent macroeconomic policy mix, with a broadly stable domestic primary fiscal deficit, helped by a sustained revenue administration effort. The external current account deficit is projected to gradually improve, as buoyant growth of exports from megaprojects in the natural resource sector is likely to outpace import growth. Stronger-than-previously-projected FDI inflows, reflecting the booming mining sector and new discoveries in the hydrocarbon sector, should continue over the medium term before tapering off in the long run.

b. **A revised public investment profile.** This reflects slower-than-previously-projected disbursements of the nonconcessional external borrowing (NCB) contracted at the time of the previous DSA, revised projections reflecting new NCB contracted recently, and the proposed increase in the NCB ceiling to US$1.5 billion under the IMF-supported program to finance projects considered critical under the authorities' public investment plan. The loans contracted under the NCB ceiling are assumed to be disbursed throughout the next four years.[12] Disbursements under the Portuguese credit line have also been updated reflecting realized disbursements.

c. **Revised grant financing projections,** reflecting a lower grant component in external financing in line with the increase of the NCB ceiling.

[11]See IMF Country Report No. 11/149.

[12]To date, authorities have signed three nonconcessional loans: (i) in December 2010, a loan from China amounting US$66 million to modernize Maputo airport; (ii) in April 2011, a loan from Brazil for the construction of Nacala airport of US$80 million, and (iii) in February 2012, a US$300 million loan from China for the construction of a ring road around Maputo.

Text Table 1. Evolution of selected macroeconomic indicators between DSA updates

Evolution of selected macroeconomic indicators

	Actual	**Preelim.**	*Projection*		
	2010	2011	2012	2013	2014
Real GDP growth (%)					
Previous DSA	6.6	7.2	7.5	7.9	7.8
Current DSA	6.8	7.1	6.7	7.2	7.8
Nominal GDP (US$ mill)					
Previous DSA	9.8	12.1	13.5	15.1	16.5
Current DSA	9.5	12.8	14.2	15.5	17.3
Primary fiscal deficit (% of GDP)					
Previous DSA	2.9	5.5	5.8	5.5	4.8
Current DSA	3.1	4.0	5.2	5.2	5.2
Current account deficit (% of GDP)					
Previous DSA	10.2	11.4	10.9	10.9	10.7
Current DSA	11.7	13.1	12.7	12.4	11.9
FDI (% of GDP)					
Previous DSA	8.1	7.2	7.1	6.5	7.0
Current DSA	8.3	16.4	11.2	10.2	10.1
Use of NCB ceiling (% of GDP)					
Previous DSA	-	2.1	2.4	2.2	1.8
Current DSA	-	1.5	2.5	3.2	3.4
Disbursements under PTL (% of GDP) 1/					
Previous DSA	1.3	1.9	1.4	1.1	0.8
Current DSA	1.4	1.8	0.9	0.9	0.5
Grant financing (% of GDP) 2/					
Previous DSA	...	9.3	9.0	8.9	8.8
Current DSA	...	9.2	8.7	8.1	8.0

1/ Portuguese credit line.
2/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new disbursements).

II. EXTERNAL DEBT SUSTAINABILITY ANALYSIS

6. Under the baseline scenario, all debt indicators remain well below their respective thresholds, including in the longer term, but external debt remains vulnerable to exchange rate volatility (Table 1 and Figure 1). The debt indicators are projected to increase over the next few years, as the authorities temporarily step up their borrowing to address the country's infrastructure gap. Stress tests reveal that Mozambique's external debt is particularly vulnerable to a one-time depreciation of the exchange rate and a negative exports shock.[13]

7. The debt sustainability indicators also seem sensitive to a recurrence of past macroeconomic performance. However, as indicated in previous DSAs, basing the analysis on a recurrence of developments of the last ten years does not take into account the significant structural changes in the Mozambican economy in the post-civil war period and the considerable improvements in the macroeconomic environment under successive Fund-supported programs, all of which have lessened the likelihood of a repeat of the past economic performance. Among other things, there has been a shift in the structure of the economy, particularly in the last five years, as large private capital inflows in the natural resources sector supported a surge in exports and a sharp fall in the current account deficit. The significant surge in capital inflows in recent years is expected to continue over the medium term, reflecting the current expansion of investment and output in the mining sector and additional gas-related discoveries which will trigger further investment and output growth over the long term.[14] Therefore, staff does not consider the historical scenario a reliable indicator of future debt distress in Mozambique as it underestimates future growth prospects and current account improvements linked to the positive impact from investment in the natural resource sector.

8. In a scenario under which non-concessional resources are not used productively, external debt indicators would deteriorate, but remain well below their respective thresholds. The customized high investment-low growth scenario assumes that foreign financing and related spending remain unchanged relative to the baseline in nominal terms. However, it assumes a much lower impact on GDP growth over the projection period from the public investment projects compared to the baseline. Although

[13] In line with the last two DSAs, the standard export shock has been tailored to capture Mozambique's true vulnerability from an export shock. This is represented by the historical volatility of prices of aluminum, which accounts for roughly half of Mozambique's export proceeds.

[14] For a more detailed explanation of the impact of different megaprojects, see IMF CR No. 11/350, Annex II.

debt burden indicators are projected to remain below the indicative thresholds, the substantial increase relative to the baseline underscores the importance of selecting projects with high economic returns, particularly when using non concessional financing.

III. PUBLIC SECTOR DEBT SUSTAINABILITY

9. The evolution of public debt indicators (including domestic debt) mirrors that of the external indicators because of the predominance of external debt (Table 3a and Figure 2). The medium-term increase in public debt reflects the temporary surge in public investment financed by external borrowing on non-concessional terms. However, over the longer term, the public debt stock projections also assume a marginal increase in domestic debt from about 2.2 percent of GDP at end-2010 to 5.7 per-cent of GDP in 2032.[7]

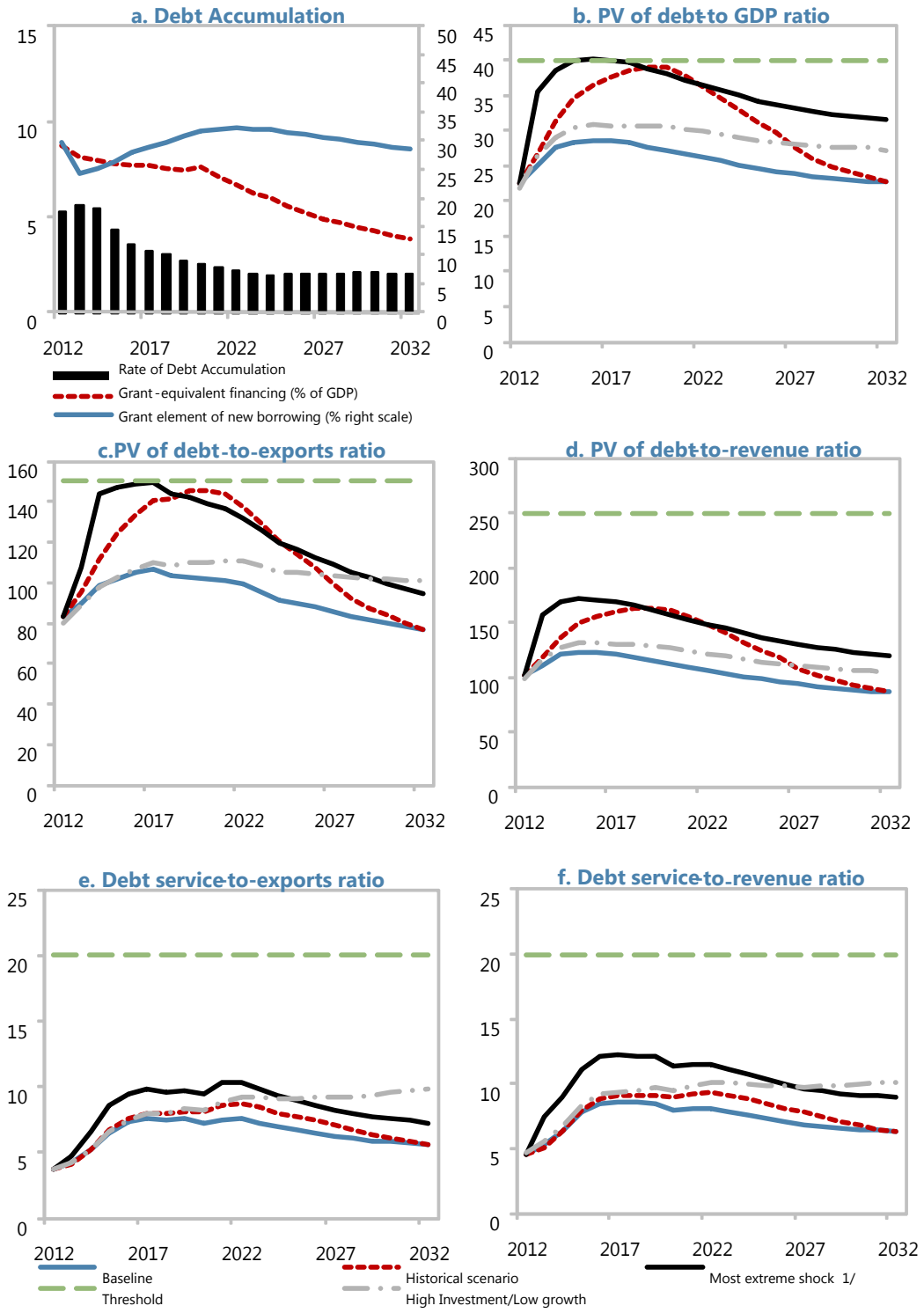
10. Over time, since the surge in public investment is expected to taper off, the primary deficit should moderate as well. However, as scenario A2 shows, continued high primary deficits could pose a risk to public debt sustainability.

IV. CONCLUSIONS

11. Mozambique's external debt dynamics continue to show a low risk of debt distress. Its external and public debt levels are expected to remain well below the respective indicative thresholds for debt distress, as the government plans to temporarily increase public investment financed by external borrowing on non-concessional terms. Nonetheless, debt vulnerability will increase, as debt ratios under stress tests and under the high-investment/low growth scenario approach the relevant thresholds. This emphasizes the need for the authorities to continue to pursue prudent macroeconomic policies and structural reforms to boost their debt management and project selection capacity, building on the initial reform steps in those areas under their IMF-supported program and those implemented with World Bank assistance.

[7]This is not expected to affect the availability of credit to the private sector.

Figure 1. Mozambique: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2012-2032[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock.

Table 1. External Debt Sustainability Framework, Baseline Scenario, 2009-2032[1]
(In percent of GDP, unless otherwise indicated)

	Actual			Historical[6/] Average	Standard[6/] Deviation	Projections									
												2012-2017			2018-2032
	2009	2010	2011			2012	2013	2014	2015	2016	2017	Average	2022	2032	Average
External debt 1/	**64.3**	**64.3**	**50.5**			**53.1**	**52.7**	**52.8**	**51.7**	**50.9**	**50.3**		**48.6**	**53.7**	
o/w public and publicly guaranteed (PPG)	38.4	39.0	31.5			38.3	40.6	43.1	43.4	43.5	43.1		39.6	33.4	
Change in external debt	-2.0	0.0	-13.9			2.7	-0.4	0.1	-1.2	-0.8	-0.6		-0.2	0.9	
Identified net debt-creating flows	3.2	6.7	-19.9			-1.6	-1.2	-1.8	-2.2	-1.5	-1.2		1.3	2.0	
Non-interest current account deficit	**10.7**	**10.1**	**11.7**	**10.7**	**3.4**	**11.2**	**10.9**	**10.4**	**9.9**	**9.6**	**9.2**		**9.3**	**9.4**	9.1
Deficit in balance of goods and services	17.4	17.8	17.3			15.8	14.9	13.9	12.9	12.3	11.7		10.3	10.6	
Exports	27.7	31.4	27.0			27.0	28.0	28.0	27.8	27.3	26.8		26.4	29.7	
Imports	45.1	49.2	44.3			42.7	42.9	42.0	40.7	39.6	38.5		36.7	40.2	
Net current transfers (negative = inflow)	-7.7	-6.9	-5.8	-7.5	2.9	-4.8	-4.5	-4.3	-4.3	-4.2	-4.0		-2.8	-1.9	-2.6
o/w official	-6.8	-6.4	-5.7			-4.6	-4.4	-4.2	-4.2	-4.1	-3.9		-2.7	-1.8	
Other current account flows (negative = net inflow)	1.0	-0.7	0.2			0.2	0.6	0.8	1.3	1.5	1.5		1.8	0.7	
Net FDI (negative = inflow)	**-8.9**	**-8.3**	**-16.4**	**-6.8**	**4.2**	**-11.2**	**-10.2**	**-10.1**	**-9.7**	**-8.8**	**-8.1**		**-6.0**	**-5.2**	-5.7
Endogenous debt dynamics 2/	**1.4**	**4.9**	**-15.2**			**-1.7**	**-2.0**	**-2.2**	**-2.3**	**-2.3**	**-2.2**		**-2.0**	**-2.1**	
Contribution from nominal interest rate	1.6	1.6	1.4			1.4	1.5	1.5	1.4	1.3	1.3		1.3	1.5	
Contribution from real GDP growth	-4.2	-4.6	-3.4			-3.1	-3.5	-3.7	-3.7	-3.6	-3.5		-3.3	-3.6	
Contribution from price and exchange rate changes	4.0	7.9	-13.1			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-5.2**	**-6.7**	**6.0**			**4.3**	**0.8**	**1.9**	**1.0**	**0.7**	**0.6**		**-1.5**	**-1.1**	
o/w exceptional financing	-4.8	-4.3	-5.4			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	36.5			37.3	37.2	37.3	36.6	36.0	35.7		35.3	42.9	
In percent of exports	...	...	135.0			138.5	132.8	133.3	131.5	132.0	133.1		133.6	144.8	
PV of PPG external debt	**...**	**...**	**17.5**			**22.5**	**25.1**	**27.6**	**28.3**	**28.6**	**28.5**		**26.2**	**22.7**	
In percent of exports	**...**	**...**	**64.7**			**83.3**	**89.5**	**98.4**	**101.8**	**104.9**	**106.3**		**99.2**	**76.5**	
In percent of government revenues	**...**	**...**	**80.0**			**101.2**	**111.2**	**120.5**	**122.1**	**121.7**	**120.6**		**106.5**	**86.0**	
Debt service-to-exports ratio (in percent)	**18.4**	**34.7**	**12.3**			**16.4**	**16.4**	**16.9**	**16.1**	**16.6**	**14.5**		**17.0**	**21.2**	
PPG debt service-to-exports ratio (in percent)	**1.7**	**2.2**	**2.2**			**3.8**	**4.2**	**5.2**	**6.5**	**7.3**	**7.6**		**7.6**	**5.7**	
PPG debt service-to-revenue ratio (in percent)	**2.6**	**3.4**	**2.8**			**4.6**	**5.2**	**6.3**	**7.8**	**8.5**	**8.6**		**8.1**	**6.4**	
Total gross financing need (Billions of U.S. dollars)	0.7	1.2	-0.2			0.6	0.8	0.9	0.9	1.1	1.2		3.2	11.3	
Non-interest current account deficit that stabilizes debt ratio	12.8	10.1	25.5			8.6	11.3	10.3	11.1	10.3	9.8		9.5	8.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.3	6.8	7.1	7.5	1.0	6.7	7.2	7.8	7.8	7.8	7.8	7.5	7.5	7.5	7.5
GDP deflator in US dollar terms (change in percent)	-5.7	-10.9	25.7	4.8	11.0	4.2	1.6	3.9	3.8	3.8	3.6	3.5	2.6	2.1	2.8
Effective interest rate (percent) 5/	2.4	2.4	2.9	3.1	1.1	3.1	3.1	3.2	3.0	2.9	2.9	3.0	3.0	3.1	3.3
Growth of exports of G&S (US dollar terms, in percent)	-14.0	8.0	15.9	14.5	13.4	10.9	13.2	11.9	11.2	9.9	9.8	11.1	10.5	11.7	11.3
Growth of imports of G&S (US dollar terms, in percent)	-2.6	3.9	21.3	14.2	10.8	7.2	9.3	9.5	8.8	8.8	8.7	8.7	10.2	11.0	10.8
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	29.8	24.4	25.2	26.2	28.1	29.0	27.1	32.2	28.5	30.7
Government revenues (excluding grants, in percent of GDP)	17.6	20.6	21.9			22.2	22.6	22.9	23.2	23.5	23.6		24.6	26.4	25.1
Aid flows (in Billions of US dollars) 7/	0.9	0.9	1.0			1.4	1.5	1.7	1.9	2.1	2.3		3.4	5.5	
o/w Grants	0.9	0.9	1.0			0.9	1.0	1.1	1.2	1.3	1.5		2.2	3.0	
o/w Concessional loans	0.0	0.0	0.0			0.5	0.5	0.6	0.7	0.8	0.8		1.2	2.4	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			8.7	8.1	8.0	7.8	7.7	7.7		6.7	3.9	5.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			63.4	59.5	59.5	62.3	64.9	66.5		68.9	58.6	65.5
Memorandum items:															
Nominal GDP (Billions of US dollars)	10.0	9.5	12.8			14.2	15.5	17.3	19.4	21.7	24.2		41.3	108.6	
Nominal dollar GDP growth	0.2	-4.9	34.6			11.3	8.9	12.0	12.0	11.9	11.7	11.3	10.3	9.7	10.5
PV of PPG external debt (in Billions of US dollars)			2.4			3.1	3.9	4.7	5.5	6.1	6.8		10.7	24.3	
(PVt-PVt-1)/GDPt-1 (in percent)						5.3	5.6	5.5	4.3	3.5	3.2	4.6	2.2	2.0	2.2
Gross workers' remittances (Billions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	17.5			22.5	25.1	27.6	28.3	28.6	28.5		26.2	22.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	64.7			83.3	89.5	98.4	101.8	104.9	106.3		99.2	76.5	
Debt service of PPG external debt (in percent of exports + remitta	...	...	2.2			3.8	4.2	5.2	6.5	7.3	7.6		7.6	5.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032
(In percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
PV of debt-to GDP ratio								
Baseline	22	25	28	28	29	29	**26**	23
A. Alternative Scenarios 1/								
A1. Key variables at their historical averages in 2012-2032 2/	22	27	31	35	36	38	**36**	23
A3. Alternative scenario: High Investment/Low growth	22	26	29	30	31	31	30	27
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	22	25	28	29	29	29	**26**	23
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	22	27	33	34	33	33	**29**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	22	27	33	34	34	34	**31**	27
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	22	31	38	38	38	37	**31**	24
B5. Combination of B1-B4 using one-half standard deviation shocks	22	30	39	39	39	38	**32**	25
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	22	35	39	40	40	40	**36**	32
PV of debt-to-exports ratio								
Baseline	83	90	98	102	105	106	**99**	77
A. Alternative Scenarios 1/								
A1. Key variables at their historical averages in 2012-2032 2/	83	95	111	124	134	140	**137**	76
A3. Alternative scenario: High Investment/Low growth	80	89	98	103	107	110	110	101
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	83	89	97	101	104	105	**97**	75
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	83	107	144	147	149	149	**132**	94
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	83	89	97	101	104	105	**97**	75
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	83	110	136	138	138	138	**118**	80
B5. Combination of B1-B4 using one-half standard deviation shocks	83	109	140	142	143	142	**123**	85
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	83	89	97	101	104	105	**97**	75
PV of debt-to-revenue ratio								
Baseline	101	111	120	122	122	121	**106**	86
A. Alternative Scenarios 1/								
A1. Key variables at their historical averages in 2012-2032 2/	101	118	136	149	155	159	**147**	86
A3. Alternative scenario: High Investment/Low growth	98	117	126	131	132	129	121	103
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	101	111	121	123	122	121	**106**	86
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	101	120	145	145	142	139	**116**	87
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	101	120	142	145	144	143	**125**	101
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	101	137	167	165	160	156	**126**	89
B5. Combination of B1-B4 using one-half standard deviation shocks	101	133	171	170	165	161	**132**	96
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	101	157	168	172	171	169	**148**	120

Table 2. Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032 (continued)
(In percent)

Debt Service-to-exports ratio

Baseline	4	4	5	7	7	8	**8**	6
A. Alternative Scenarios 1/								
A1. Key variables at their historical averages in 2012-2032 2/	4	4	5	7	8	8	**9**	6
A3. Alternative scenario: High Investment/Low growth	4	4	5	7	7	8	9	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	4	4	5	7	7	8	**8**	6
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	4	5	7	9	10	10	**10**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	4	4	5	7	7	8	**8**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	4	4	6	7	8	8	**9**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	6	8	9	9	**10**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	4	4	5	7	7	8	**8**	6

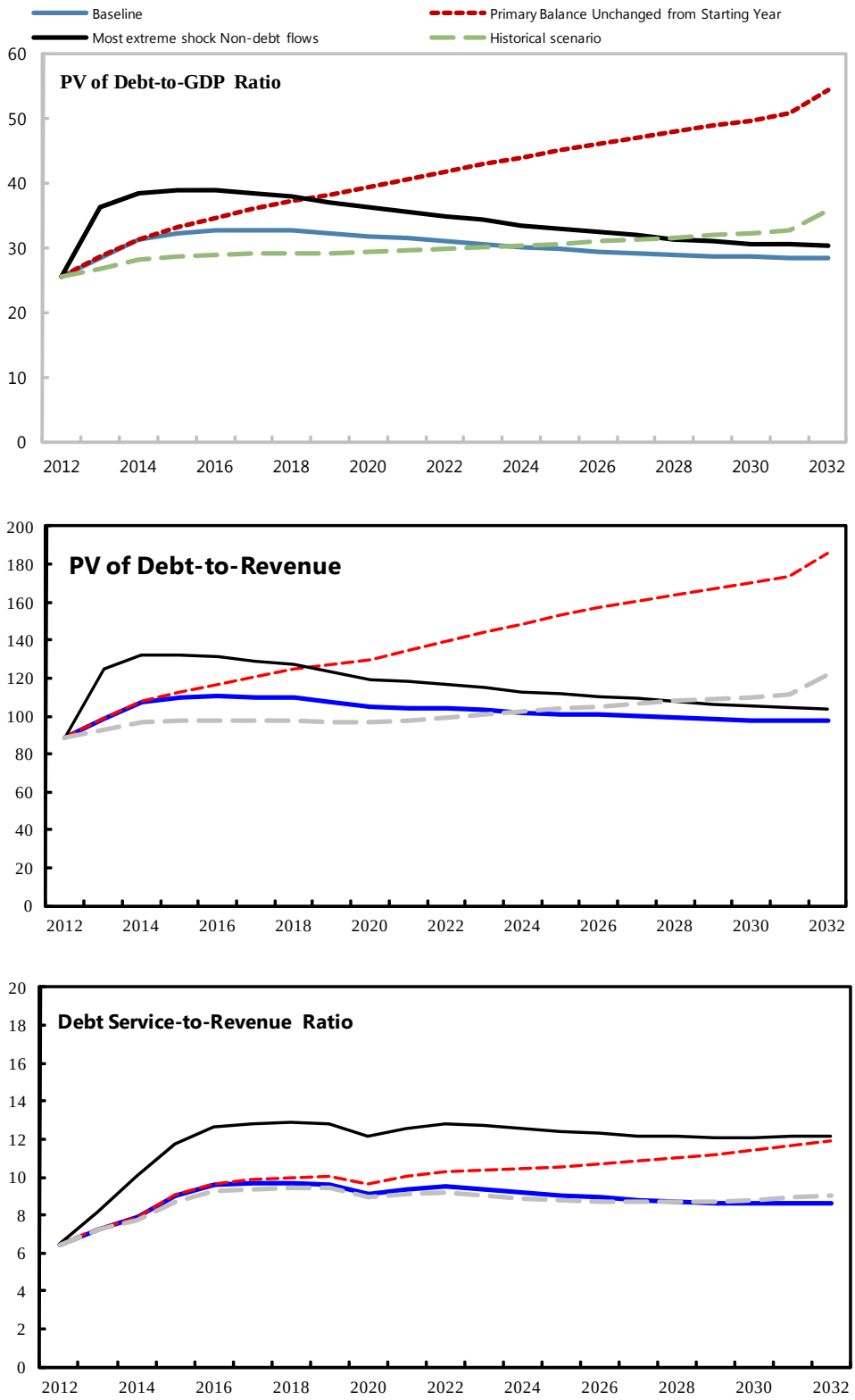
Debt service-to-revenue ratio

Baseline	5	5	6	8	9	9	**8**	6
A. Alternative Scenarios 1/								
A1. Key variables at their historical averages in 2012-2032 2/	5	5	6	8	9	9	**9**	6
A3. Alternative scenario: High Investment/Low growth	5	6	7	8	9	9	10	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	5	5	6	8	9	9	**8**	6
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	5	5	7	8	9	9	**9**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	5	6	8	9	10	10	**10**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	5	5	7	9	10	10	**10**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	7	9	10	10	**10**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	5	7	9	11	12	12	**12**	9
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	22	22	22	22	22	22	**22**	22

Sources: Country authorities; and staff estimates and projections.

1/ The stress test A2 (borrowing in less concessional terms) has not been included in this case (in line with the 2010 and 2011 DSAs). Given the commercial financing terms already included in the baseline for the investment projects, this scenario would yield unrealistic results, as marginal borrowing under this shock is calculated on the average terms of new borrowing; therefore assuming harsher terms than Mozambique would actually face in the need to cover a financing gap.
2/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 2. Mozambique: Indicators of Public of Public Debt Under Alternative Scenarios, 2012-2032[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2022.
2/ Revenues are defined inclusive of grants.

Table 3a. Mozambique: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009-2032
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2009	2010	2011	Average 5/	Standard Deviation 5/	2012	2013	2014	2015	2016	2017	2012-17 Average	2022	2032	2018-32 Average
Public sector debt 1/	41.6	41.4	34.4			41.4	43.9	46.7	47.2	47.5	47.3		44.4	39.1	
o/w foreign-currency denominated	38.4	39.0	31.5			38.3	40.6	43.1	43.4	43.5	43.1		39.6	33.4	
Change in public sector debt	-0.5	-0.2	-7.0			7.0	2.6	2.8	0.5	0.3	-0.2		-0.5	-0.3	
Identified debt-creating flows	4.5	3.9	-7.3			6.1	1.9	2.2	0.3	0.4	-0.1		-0.4	-3.6	
Primary deficit	5.0	3.1	4.0	3.1	1.0	5.2	5.2	5.2	4.2	3.8	3.4	4.5	2.5	-1.3	2.2
Revenue and grants	27.1	29.7	29.6			28.8	28.9	29.0	29.3	29.6	29.8		29.8	29.2	
of which: grants	9.5	9.1	7.7			6.6	6.3	6.1	6.1	6.1	6.1		5.2	2.8	
Primary (noninterest) expenditure	32.1	32.8	33.6			34.0	34.2	34.2	33.5	33.4	33.2		32.3	27.8	
Automatic debt dynamics	-0.3	1.0	-11.3			0.8	-3.4	-3.0	-3.9	-3.4	-3.5		-2.9	-2.3	
Contribution from interest rate/growth differential	0.0	2.3	-9.5			-2.4	-2.2	-3.3	-3.6	-3.6	-3.5		-2.9	-2.3	
of which: contribution from average real interest rate	2.5	5.0	-6.7			-0.2	0.6	-0.2	-0.2	-0.2	-0.1		0.3	0.4	
of which: contribution from real GDP growth	-2.5	-2.6	-2.8			-2.2	-2.8	-3.2	-3.4	-3.4	-3.4		-3.1	-2.7	
Contribution from real exchange rate depreciation	-0.3	-1.4	-1.8			3.2	-1.2	0.4	-0.3	0.2	0.0		...	...	
Other identified debt-creating flows	-0.3	-0.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.2	-0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.1	-0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-5.0	-4.1	0.2			0.9	0.7	0.5	0.3	-0.1	-0.1		-0.2	3.4	
Other Sustainability Indicators															
PV of public sector debt	...	...	20.4			25.6	28.5	31.2	32.1	32.6	32.7		31.0	28.3	
o/w foreign-currency denominated	...	...	17.5			22.5	25.1	27.6	28.3	28.6	28.5		26.2	22.7	
o/w external	...	...	17.5			22.5	25.1	27.6	28.3	28.6	28.5		26.2	22.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	5.9	4.7	5.4			7.1	7.3	7.5	6.8	6.6	6.3		5.3	1.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	68.9			88.8	98.4	107.4	109.7	110.4	109.9		103.9	97.2	
PV of public sector debt-to-revenue ratio (in percent)	…	…	93.3			115.2	126.0	136.2	138.6	138.9	138.4		126.1	107.5	
o/w external 3/	…	…	80.0			101.2	111.2	120.5	122.1	121.7	120.6		106.5	86.0	
Debt service-to-revenue and grants ratio (in percent) 4/	3.5	5.1	4.6			6.5	7.3	7.9	9.0	9.6	9.7		9.5	8.6	
Debt service-to-revenue ratio (in percent) 4/	5.4	7.4	6.3			8.4	9.3	10.0	11.4	12.1	12.2		11.5	9.5	
Primary deficit that stabilizes the debt-to-GDP ratio	5.5	3.3	11.1			-1.8	2.6	2.4	3.6	3.5	3.6		3.0	-1.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.3	6.8	7.1	7.5	1.0	6.7	7.2	7.8	7.8	7.8	7.8	7.5	7.5	7.5	7.5
Average nominal interest rate on forex debt (in percent)	0.6	0.6	0.7	0.6	0.3	1.2	1.4	1.6	1.7	1.8	1.8	1.6	1.6	1.6	1.6
Average real interest rate on domestic debt (in percent)	3.9	10.5	22.0	12.5	16.0	23.7	23.1	21.1	19.3	18.0	16.8	20.3	13.8	11.0	12.7
Real exchange rate depreciation (in percent, + indicates depreciati	-0.8	-3.4	-6.2	-1.2	12.0	11.3	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.2	10.0	10.7	7.9	2.1	7.6	6.0	6.0	6.0	5.9	5.8	6.2	5.8	5.2	5.7
Growth of real primary spending (deflated by GDP deflator, in perc	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	29.8	24.4	25.2	26.2	28.1	29.0	27.1	32.2	28.5	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3b. Mozambique: Sensitivity Analysis for Key Indicators of Public Debt, 2012-2032

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
PV of Debt-to-GDP Ratio								
Baseline	26	28	31	32	33	33	31	28
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	26	27	28	29	29	29	30	36
A2. Primary balance is unchanged from 2012	26	28	31	33	35	36	42	54
A3. Permanently lower GDP growth 1/	26	29	31	33	33	34	33	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	26	29	32	33	34	35	34	33
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	26	28	29	31	31	31	30	28
B3. Combination of B1-B2 using one half standard deviation shocks	26	27	29	30	31	32	31	30
B4. One-time 30 percent real depreciation in 2013	26	37	38	38	38	37	33	31
B5. 10 percent of GDP increase in other debt-creating flows in 2013	26	36	38	39	39	38	35	30
PV of Debt-to-Revenue Ratio 2/								
Baseline	89	98	107	110	110	110	104	97
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	89	92	97	97	97	97	99	122
A2. Primary balance is unchanged from 2012	89	98	108	113	117	121	140	186
A3. Permanently lower GDP growth 1/	89	99	108	111	112	112	111	121
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	89	99	110	114	115	116	113	112
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	89	95	102	104	106	105	101	96
B3. Combination of B1-B2 using one half standard deviation shocks	89	94	100	103	105	106	104	102
B4. One-time 30 percent real depreciation in 2013	89	127	132	130	127	124	112	106
B5. 10 percent of GDP increase in other debt-creating flows in 2013	89	125	132	132	131	129	117	104
Debt Service-to-Revenue Ratio 2/								
Baseline	6	7	8	9	10	10	9	9
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	6	7	8	9	9	9	9	9
A2. Primary balance is unchanged from 2012	6	7	8	9	10	10	10	12
A3. Permanently lower GDP growth 1/	6	7	8	9	10	10	10	10
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	6	7	8	9	10	10	10	9
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	6	7	8	9	9	10	9	8
B3. Combination of B1-B2 using one half standard deviation shocks	6	7	8	9	9	10	9	9
B4. One-time 30 percent real depreciation in 2013	6	8	10	12	13	13	13	12
B5. 10 percent of GDP increase in other debt-creating flows in 2013	6	7	9	10	10	10	10	9

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.